LESSON 10.3

Math is Playful

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use patterns and relationships to make sense of the Tower of Hanoi puzzle and think logically about its solutions.

WHY THIS MATTERS The Tower of Hanoi is a puzzle invented in 1883. It has three rods and a stack of discs, smallest on top to largest on the bottom. The number of discs can vary, but the number of rods does not — and hiding inside it is one of the prettiest patterns in mathematics.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
puzzle *Spanish: rompecabezas*	A problem designed to be played with — it has rules, a goal, and room to imagine solutions.	Tower of Hanoi: three rods, a stack of discs, and two rules.
pattern *Spanish: patrón*	Something that repeats or changes in a predictable way.	Circle, square, triangle, circle, square, triangle …
pattern rule *Spanish: regla del patrón*	A rule that tells you how to get the next value in a pattern from the one before it.	1 → 3 → 7 → 15: the rule is “double it and add 1”.
predict *Spanish: predecir*	To say what will happen before it happens, using a pattern or reasoning.	We counted towers of 1, 2 and 3 discs. Reach PAST them: the rule says 5 discs takes 31 steps — before moving one.
organize *Spanish: organizar*	To arrange information — often in a table — so patterns become easier to see.	Discs 1, 2, 3 in one column and steps 1, 3, 7 in the next.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Before moving anything, I describe what the solved puzzle will look like: the discs in the same order — smallest on top to largest on bottom — but on a different rod.

1. Then I play the small cases: a one-disc tower takes 1 move, and a two-disc tower takes 3 moves.
2. A three-disc tower takes at least 7 moves. Small cases are how mathematicians play.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Discs: 1, 2, 3 — Steps: 1, 3, 7. What is happening from each row to the next?

1. _____
2. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The rule is _____, so _____ is legal / breaks it because _____.
- The number of _____ never changes.
- There are always _____ .
- Being playful with math means _____ .

Word bank puzzle | pattern | pattern rule | predict | organize

WATCH OUT!

A common mistake

Doubling the previous number of steps but forgetting to add 1

LESSON 10.3

Math is Playful

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

What is the fewest number of moves needed to solve a three-disc Tower of Hanoi puzzle?

(A) 3 moves
(B) 6 moves
(C) 7 moves
(D) 9 moves

HINT It is more than one move per disc.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which move is against the rules of the Tower of Hanoi?

(A) Moving one disc at a time
(B) Moving the top disc from a stack
(C) Placing a larger disc on top of a smaller disc
(D) Placing a smaller disc on top of a larger disc

HINT Two of these are the rules themselves.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What will the solved puzzle look like?

(A) The discs in the same order — smallest on top to largest on bottom — but on a different rod

(B) The discs in reverse order, largest on top

(C) The discs spread out with one disc on each rod

(D) The discs back on the starting rod in any order

HINT Imagine the ending before you start.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

How many moves does a one-disc tower take? A two-disc tower?

(A) 1 move and 3 moves

(B) 1 move and 2 moves

(C) 1 move and 4 moves

(D) 2 moves and 4 moves

HINT Play the one-disc game in your head first.

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

The number of discs in a Tower of Hanoi puzzle can vary. What NEVER varies?

SENTENCE STARTER The number of ___ never changes.
There are always ___ .

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

What does it mean to be playful with math? Use the Tower of Hanoi as your example.

SENTENCE STARTER Being playful with math means ___ .
With the Tower of Hanoi, I ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Take a move you called legal and your partner called a rule-break. What is the rule in your own words, and does it still hold on a move you both agree about?

SENTENCE STARTER The rule is ___, so ___ is legal / breaks it because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 10.3

Math is Playful

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

The table shows 1 disc → 1 step, 2 discs → 3 steps, 3 discs → 7 steps. What is the pattern rule for the number of steps when you add one more disc?

(A) Add 2 to the number of steps
(B) Add the number of discs to the steps
(C) Multiply the number of steps by 3
(D) Double the number of steps and add 1

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Use the pattern rule to complete the table: how many steps for a four-disc tower?

(A) 9 steps
(B) 14 steps
(C) 15 steps
(D) 28 steps

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is the minimum number of steps to solve a FIVE-disc Tower of Hanoi puzzle?

(A) 15 steps
(B) 30 steps
(C) 31 steps
(D) 35 steps

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Why is finding a pattern MORE powerful than just solving the three-disc puzzle by hand?

(A) The pattern lets you skip the rules of the puzzle
(B) The pattern predicts the answer for towers you have never played, like five or six discs
(C) Patterns make the discs move by themselves
(D) A pattern proves the puzzle is impossible

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

How can you ORGANIZE the information about discs and moves to help you see patterns?

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

Before solving, the lesson asked you to DESCRIBE what the solution will look like. Why is imagining the ending a useful way to start a puzzle?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Take a move you called legal and your partner called a rule-break. What is the rule in your own words, and does it still hold on a move you both agree about?

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LESSON 10.3

Math is Playful

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each tower to the calculation that gives its fewest moves.

_____ 4-disc tower

_____ 5-disc tower

_____ 6-disc tower

_____ Any n-disc tower

(A) $2 \times 7 + 1 = 15$

(B) $2 \times 15 + 1 = 31$

(C) $2 \times 31 + 1 = 63$

(D) Double the $(n - 1)$-disc answer, add 1

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A five-disc tower takes 31 steps. Using the pattern rule, how many steps would a SIX-disc tower take?

(A) 32 steps

(B) 36 steps

(C) 62 steps

(D) 63 steps

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Look at the table a different way: 1 disc → 1, 2 discs → 3, 3 discs → 7, 4 discs → 15, 5 discs → 31. Which describes the relationship between the number of discs and the number of steps?

(A) The steps are the number of discs multiplied by itself

(B) The steps are 6 times the number of discs, plus or minus 1

(C) The steps are the two previous step counts added together

(D) The steps are 1 less than a doubling pattern: 2, 4, 8, 16, 32 each minus 1

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Answer in complete sentences.**

The 15 Puzzle is a sliding puzzle with 15 tiles in a frame 4 tiles high by 4 tiles wide. You arrange the tiles from 1 to 15 by sliding one tile at a time into the open space. Sometimes it is solvable and sometimes it is not. How would you make an argument about whether a particular 15 Puzzle is solvable, using what this lesson taught about being playful?

WRITE YOUR ANSWER

5 **Answer in complete sentences.**

The summary says playing with math lets us think differently by looking for patterns AND by imagining ways to visualize a problem. Describe how you used BOTH of those with the Tower of Hanoi.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Take a move you called legal and your partner called a rule-break. What is the rule in your own words, and does it still hold on a move you both agree about?

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